

LESSON GUIDE

Technology Plus

Unit 3: Hardware

2.2.1 – Internal Computing Components

Lesson Overview:

In this lesson, students will explore the internal components of computing devices, identify their purposes, and how they compare to parts of the human body.

Students will:

- Identify internal computing components.
- Describe the purpose of internal computing components.
- Explain how internal computing components compare to parts of the human body.

Guiding Question: What are the internal components of computing devices and what are their purposes?

Suggested Grade Levels: 8-12

Technology Needed: Yes

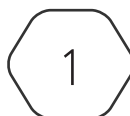
Approximate Time Required: 70-95 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 2.2 – Explain the purpose of common internal computing components.
 - o Motherboard/system board
 - o Firmware/basic input/output system (BIOS)
 - o Random-access memory (RAM)
 - o Central processing unit (CPU)
 - o Graphics processing unit (GPU)
 - o Storage
 - o Network interface card (NIC)
 - Wired vs. wireless
 - Onboard vs. expansion card

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Lesson Background

Background Information:

In this lesson, students will learn about the internal components that make up computing devices and explore their purposes. Each internal component is crucial to the successful function of a computer and all components work together to provide a smooth user-friendly experience. Students will learn the different components, the functions and operations of each component, and how they relate to the human body. Just like we need our organs to function properly for survival, a computer needs internal components to process, store, and communicate data throughout its system. Students will also use Windows and Linux system tools to explore the internal computing components of virtual machines – which can be applied to identify the internal components of their physical computers.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 2.2.1 – Internal Computing Components • Lab Slides 2.2.1 – CSI Computer System Investigation
Materials per Student	• Student Notes 2.2.1 – Internal Computing Components • Student Handout 2.2.1 – Internal Computing Components • Assessment 2.2.1 – Internal Computing Components (optional)

Cyber Terms:

Term	Definition
motherboard	the main circuit board that connects and allows communication between all other internal components
basic input/output system (BIOS)	software embedded in a computer's motherboard that initiates at the beginning of the boot process
random access memory (RAM)	a system's short-term memory that provides the temporary storage a CPU needs to perform tasks quickly
central processing unit (CPU)	responsible for executing instructions, processing data and coordinating the computer's operations
graphics processing unit (GPU)	a visual system that allows a user to interact with an operating system using icons, buttons, and menus instead of text-based commands
storage	mechanisms and devices used to save and retain data and programs for later use; saving output short-term (in system memory) or long-term (on hard drives)
network interface card (NIC)	enables a computer to connect to a network so it can communicate with other devices and access network resources

2.2.1 – Internal Computing Components

Guiding Question: What are the internal components of computing devices and what are their purposes?

Lesson Launch (5 minutes)

- **Ask:** What do you think would happen if your brain couldn't send signals, your memory was wiped every time you blinked, or your organs had no way to connect with each other?
- Allow students to share their answers.
- **Say:** Just like us, computers have internal components that are like parts of the human body. Without specific components, computers would not be able to make decisions, store data, communicate, or interact with other devices.
- **Say:** Today, we will explore the internal computing components, their purposes, and how they compare to parts of the human body.
- Take this time to introduce students to the abbreviations they will see or hear frequently during the lesson and what they stand for.
 - BIOS – basic input/output system
 - RAM – random-access memory
 - CPU – central processing unit
 - GPU – graphics processing unit
 - NIC – network interface card

Hardware Blueprint: Inside a Computer (30-40 minutes)

- Distribute Student Handout 2.2.1 – Internal Computing Devices and encourage students to fill in the answers during the lesson.

Motherboard

- Introduce the motherboard as the most important component of a computer's internal structure.
- Share the definition of a motherboard and discuss the components of a motherboard in detail.
- Motherboard Components:
 - Form factors are the design of a motherboard – form factors fit into two main categories: proprietary and standardized.
 - A proprietary motherboard is made to fit a specific system, like a laptop brand or cellphone.
 - A standardized motherboard is not made for a specific system.

- The motherboard is the central hub of a computer because it connects all other internal computing components and allows them to communicate with each other.
- The motherboard has electrical pathways that facilitate the transfer of data and power between different components. These pathways ensure signals can travel between peripheral devices – connected for functionality – such as a mouse, keyboard, extended displays, and more.
- Chipsets control the flow of data between the processor, memory, and other hardware – divided into two parts: northbridge and southbridge.
 - Northbridge manages high-speed connections like CPU and RAM.
 - Southbridge manages slower connections like USB ports and hard drives.
- Motherboards come with expansion slots that allow for upgrades and added functionality. Common types of expansion slots include:
 - PCIe – peripheral component interconnect express, used for GPUs, NICs, sound cards, and storage expansion.
 - M.2 – used for storage solid-state drives, and sometimes Wi-Fi and Bluetooth cards.
 - DIMM – dual inline memory module, used for adding RAM.
- BIOS is firmware installed on the motherboard that boots the system and initiates the memory and hard drive allowing the OS to start.
- The motherboard distributes power from the power supply unit (PSU) to other internal components including peripherals.
- I/O ports are included for connecting any peripheral devices to their computer.
- The motherboard is like the nervous system of the computer because it connects all other components together and allows them to communicate.

Firmware/basic input/output system (BIOS)

- Introduce **BIOS** as firmware - which is any software embedded directly into hardware devices to control their basic functions and operations – embedded in a computer's motherboard. It is typically stored on a read-only memory (ROM) chip.
- Share the definition of BIOS and discuss the functions of BIOS in detail.
- Functions of BIOS:
 - When a computer is powered on, BIOS initiates power-on self-test (POST) to check if the fundamental hardware components and storage devices are functioning correctly.
 - BIOS initializes and configures the settings of hardware components so they can be utilized by the OS.
 - After POST and the hardware initialization, BIOS finds the boot storage device, loads the boot into memory, and then loads the boot to the OS. The OS continues the startup.
 - BIOS comes with a configuration utility for users to configure their hardware settings.
 - Do a web search to find specific instructions on accessing BIOS setup on a device.
 - BIOS translates OS requests into commands the hardware can understand and execute.
 - Most modern OS come with their own drivers, but BIOS includes basic drivers for functions like

keyboard input and display output.

- BIOS manages power settings and sleep states ensuring the computer uses power efficiently and can wake up from sleep modes correctly.
- BIOS can be upgraded by flashing the code on its ROM chip, it is non-volatile and keeps its settings when the computer is turned off. Upgrading BIOS can help the motherboard function better.
- Discuss Unified Extensible Firmware Interface (UEFI) and how it is used in modern computers - taking the place of BIOS. The terms are used interchangeably and devices with UEFI may still be referred to as using BIOS.
- BIOS is like the internal reflexes of the computer because it completes the basic start-up and low-level functions automatically.

Random-Access Memory (RAM)

- Introduce **RAM** as a crucial component in a computer's architecture - the capacity and speed of RAM impacts a system's performance.
- Share the definition of RAM and discuss the types of RAM briefly.
- Types of RAM:
 - Dynamic RAM (DRAM) is used in modern computers and stores each bit of data into a separate capacitor within an integrated circuit.
 - DRAM needs to be refreshed thousands of times per second to retain data.
 - Mention Dynamic RAM is also known as volatile memory - memory that loses its data when a system is powered off.
 - Static RAM (SRAM) is faster and more expensive than DRAM because it does not need to be refreshed.
 - SRAM is primarily used for CPU cache. Cache is discussed more in depth under the CPU section.
 - Mention Static RAM is also known as non-volatile memory - memory that retains its data even when a system is powered off.
 - Double Data Rate RAM (DDR RAM) is a type of DRAM that transfers data twice per clock cycle doubling the data transfer rate.
 - Most DRAM used today is DDR RAM, some versions of DDR RAM include DDR3, DDR4, and DDR5. Each new version increases in speed and efficiency.
- Discuss the functions and operations of RAM in detail. Some of the bullets left blank have all details needed on the slides, while others include elaborate details that are not on the slides.
 - Working Memory: When running a program, the computer loads instructions into the RAM and the CPU takes the first line of instructions, does what it says, and writes the result back into the memory.
 - Fast Access
 - Handling Multiple Tasks: Having a lot of RAM reduces the possibility of the system slowing down from handling multiple tasks at once.
 - More RAM = More Power
 - Volatile Memory

- Random Access
- DIMMs:
 - Small outline DIMMs (SODIMM) are made for laptops or smaller computers, with lower power usage, and shorter RAM slots.
 - DIMMs are used for desktops, workstations, and servers with higher power usage and larger RAM slots.
- RAM acts as the computer's short-term memory and provides the temporary storage a CPU needs to perform tasks quickly.

Central Processing Unit (CPU)

- Introduce the **CPU** as the most important component on the motherboard.
- RAM holds memory for CPU and executes one line at a time
- Share the definition of a CPU, the two main categories of CPUs, and discuss the types of CPUs briefly.
- Types of CPUs:
 - Desktop CPUs: Designed for personal computers and workstations, they have higher clock speeds and more cores to handle demanding tasks. Utilized for high performance.
 - Mobile CPUs: Used in laptops, tablets, and smartphones, they are optimized for lower power consumption and longer battery life. Utilized for power efficiency.
 - Server CPUs: Built for servers and data centers, with features like additional cores, larger caches, and support for larger amounts of memory. Utilized for scalability and reliability.
 - Embedded CPUs: Found in embedded systems like appliances, vehicles, and industrial machines, designed for specific tasks and environments. Utilized for specialized applications.
- Discuss the functions and operations of CPUs in detail.
 - Processing Commands
 - Manipulating Data
 - Managing Operations
 - Arithmetic Logic Unit (ALU): Responsible for the actual computation - or for calculating the result or output of mathematical and logical operations.
 - Control Unit (CU): Manages the flow of data between the CPU and other hardware components. CU is responsible for performing the operations - or carrying out any instructions.
 - Registers: Hold the data a CPU is processing in real-time, while the RAM is holding the data waiting to be processed.
 - Instruction Cycle
 - Clock Cycle: Higher clock speeds usually mean faster processing.
 - Multi-Core Processors: More cores usually mean more computer power. Cores range from 4, 8, and 16.
 - Cache Memory: Cache helps the CPU work faster.
 - Cache vs. RAM: RAM is between the CPU cache and the storage drives. RAM is faster than

storage, but slower than CPU cache. CPU cache is used for the most frequently accessed data.

- The CPU is like the computer's brain, it is responsible for executing instructions, processing data, and coordinating the computer's operations.

Graphics Processing Unit (GPU)

- Introduce the **GPU**, also known as a graphics card. GPUs can be built into the CPU of a computer used for basic tasks, while computers that need powerful graphics will have a GPU added.
- Share the definition for a GPU and discuss the types of GPUs briefly.
- Types of GPUs:
 - Integrated GPUs: Built into the CPU or motherboard and shares system memory with the CPU, suitable for basic tasks and light gaming. These have a lowered performance compared to dedicated GPUs.
 - Dedicated GPUs: Standalone GPUs with their own VRAM and processing power, suitable for gaming, professional work, and high-performance computing.
 - Mobile GPUs: Found in laptops and mobile devices, designed to balance performance and power efficiency.
 - Workstation GPUs: Designed for professional applications that need high precision and error correction for tasks like 3D modeling, video editing, and simulations.
 - Compute GPUs: Designed for general-purpose computing tasks, specifically those used in data centers and for machine learning applications.
- Discuss the functions and operations of GPUs in detail.
 - Image and Video Processing: Generates pixels and textures to display images on a screen.
 - Parallel Processing
 - General-Purpose GPU (GPGPU)
 - Numerous Cores: Unlike CPUs which only have a few cores and a general purpose for sequential processing, GPUs are highly specialized and built for parallel processing.
 - Stages of Processing
 - Video RAM (VRAM)
 - Programming Effects
- Discuss resolution and frame rate briefly - resolution is how clear a picture looks, and frame rate is how smooth the motion looks.
- The GPU is like the computer's eyes because it is responsible for displaying the user interface of a computer and performing parallel processing tasks.

Storage

- Introduce **storage** - without storage devices a computer would not be able to remember anything or run any programs. It would be like having a brain with no memory.
- Share the definition of storage and discuss its key characteristics in detail.

- Key Characteristics of Storage:
 - Capacity: Storage capacity is measured by how much data the device can hold.
 - Performance: Storage performance is determined by how quickly data can be read from or written to the device.
 - Reliability: Reliability is the likelihood of a device failing and its ability to preserve data integrity.
 - Form Factor: Storage devices come in various physical sizes and shapes, which are determined based on the requirements of the computer or storage system.
- Discuss the types of storage briefly.
 - RAM is used for temporary storage of data that is actively being used or processed. RAM is volatile, losing all its contents when powered off.
 - Secondary storage types are often built-in, but a user can upgrade them.
 - Tertiary storage types are often optional and added by the user.
 - Network storage keeps data somewhere on the network instead of a user's computer.
 - Secondary and tertiary storage types are non-volatile because they do not lose data every time the system shuts down.
- Storage is the computer's long-term memory because it holds all the software, files, and data necessary for successful operation and execution.
- Students will learn more about secondary and tertiary storage in lesson 2.3.1, and network storage in lesson 2.3.2.

Network Interface Card (NIC)

- Introduce **NICs**, sharing their definition, and their different forms.
- Discuss Wired vs. Wireless NICs in detail.
 - Wired NICs are commonly used in desktops, servers, and network switches.
 - Common Ethernet standards for wired NICs include 100 Mbps, 1000 Mbps, and 10Gbps.
 - Wireless standards include IEEE 802.11a/b/g/n/ac/ax – each type (separated by /).
- Discuss Onboard vs. Expansion NICs in detail.
 - Most modern motherboards come with one or more integrated NICs.
 - Redundancy is having multiple systems or NICs in place for availability and failover will switch to the backup system or NIC in the event of failure.
- The NIC is like the computer's mouth and ears because it enables a computer to connect to a network so it can communicate with other devices and access network resources.

Cooling Systems

- Introduce cooling systems, their definition, and the main cooling types.
- Cooling allows everything to keep running efficiently.
- Air cooling is the most common type of cooling.

- Liquid cooling is typically used in gaming PCs and servers.
- Heat sinks are sometimes combined with fans.
- The cooling system is like the computer's temperature control, helping the system stay cool to keep everything safe and working properly.
- Mention that a device must have a motherboard, RAM, and a CPU to be considered a computing device, all other components are optional enhancements to the functionality of a computer.

Lab : CSI: Computer System Investigation (25-35 minutes)

- This lab requires access to the CYBER.ORG Range, the Ubuntu Linux machine, the Windows Server 22 machine, and Lab Slides 2.2.1– CSI Computer System Investigation.
- Introduce the guiding question and objectives for the lab.
- Students will use Ubuntu and Windows VMs to explore internal computing components using embedded system tools.
- Discuss lab key terms.
- Students should open two separate tabs in their browser for each virtual machine. This will allow them to easily switch back and forth during the lab.
- Students should complete the lab portion of their student handout.
- Student answers may slightly vary from the handout answer - but should be in a similar format.
- Students will notice similarities between the machines and should take away the differences in how the system information is gathered from each machine.
- Similarities occur because they are using virtual machines, and the hardware is simulated making components masked or generic.
- Windows Server 22: Instruct half of the class to keep applications open after retrieving system information, instruct the other half of the class to close the applications. When viewing real-time CPU, RAM, and Storage - have students compare their results.

Helpful Tips

- When viewing real-time activity, the data will fluctuate. Encourage students to observe and write the average for their answers – it won't be an exact measurement and will vary per student.
- Students will see GiB and MiB when viewing storage space in the Ubuntu machine.
 - GB or MB is in decimal (base 10) format.
 - 1 GB = 1,000,000,000 bytes
 - GiB or MiB is in binary (base 2) format.
 - 1GiB = 1,073,741,824 bytes
 - Students are not required to convert binary to decimal in the lab.
 - Peripheral Component Interconnect (PCI) Devices
 - VGA Amazon device does not give specific model; students should just name Amazon Device #####

Putting It All Together

- What similarities in output did the two machines have during the lab?

Some of the internal components were the same such as BIOS, CPU, and Motherboard manufacturers.

- Why do you think these similarities occur?

I think these similarities occur because virtual machines simulate their hardware and can be identical unless settings are changed.

- What differences in output did you notice between the two machines?

The Windows machine had a simpler output and was user friendly. The Ubuntu machine required more steps to find the system information and the output requires the user to carefully look over the terminal.

Putting It All Together (10-15 minutes)

Discussion Questions

- Review what students learned today by answering the questions below as a class.
- Why are the motherboard and CPU so important?

The motherboard is what connects and allows communication between all internal components of a computer, and a CPU is responsible for executing instructions, processing data, and coordinating the computer's operations. Without a motherboard, the computer would just be a pile of parts that could not connect or communicate with each other to function. Without a CPU, the computer would not be able to run software, load the OS, or process any data.

- What is the difference between RAM and storage?

RAM acts as the system's short-term memory and provides the temporary storage a CPU needs to perform tasks quickly. Storage holds all the software, files, and data necessary for the operation of a system and the execution of applications. RAM is volatile and deletes its data when the system is powered off; storage is non-volatile and keeps its data when the system is powered off.

- How does BIOS positively impact a computer?

BIOS ensures that the hardware and storage of a computer are functioning correctly during the boot process, and it is responsible for loading the boot into memory and the OS, so the OS can take over and complete the startup of a system.

- What are the main functions of GPUs?

The main functions of GPUs are image and video processing, parallel processing, and providing general-purpose GPUs for computing tasks.

- What should a user consider when choosing a NIC and why?

When choosing a NIC a user should consider their performance requirements, physical space, and budget because some NICs have higher transfer speeds, some are costly and take up more space, and some are more secure. A user should consider their needs and choose accordingly.